



AU-Conditioned MAE + Pose-Normalized FER

Robust Facial Expression Recognition with AU-MAE and Pose Normalization

Team Members

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Key Roles

- Pretraining (AU-MAE, backbone models, GitHub pipeline)
- Pose normalization, datasets & evaluation, ablation studies

The Challenge: Robust FER In-the-Wild

Facial Expression Recognition (FER) is a cornerstone technology, driving innovation across Human-Computer Interaction (HCI), healthcare, and robotics. Its broad applicability underscores the critical need for systems that can accurately interpret human emotion.



Current Limitations in FER

- **Large Pose Variations:** Significant performance drop between frontal and profile views.
- **Occlusions:** Masks, sunglasses, or hands obscuring facial features.
- **Limited Labeled Data:** Scarcity of high-quality, comprehensively annotated datasets.
- **Subtle Expressions & Imbalance:** Difficulty in discerning nuanced emotions and handling skewed class distributions.

Our goal is to develop a **robust, interpretable FER system** capable of accurately recognizing expressions from "in-the-wild" images and videos.

Related Work & The Gap

Datasets

Major benchmarks like AffectNet, RAF-DB, and FER-2013 provide foundational data, but often struggle with the diversity of real-world conditions.

Pretraining

Self-supervised MAE pretraining has shown promise for general face tasks, learning robust representations from unlabeled data.

The Unaddressed Need

Few existing works effectively combine **AU-guided self-supervised pretraining** with **explicit pose normalization** for enhanced FER robustness in unconstrained environments.

Architectures

CNNs with attention excel at local feature extraction (eyes, mouth).

Vision Transformers (ViT, VTFF) capture global contextual information.

Multi-task Learning

Some models integrate **Action Unit (AU) detection** with FER, but deep integration within self-supervised pretraining remains underexplored.

Our Novel Contributions

AU-Conditioned MAE Pretraining

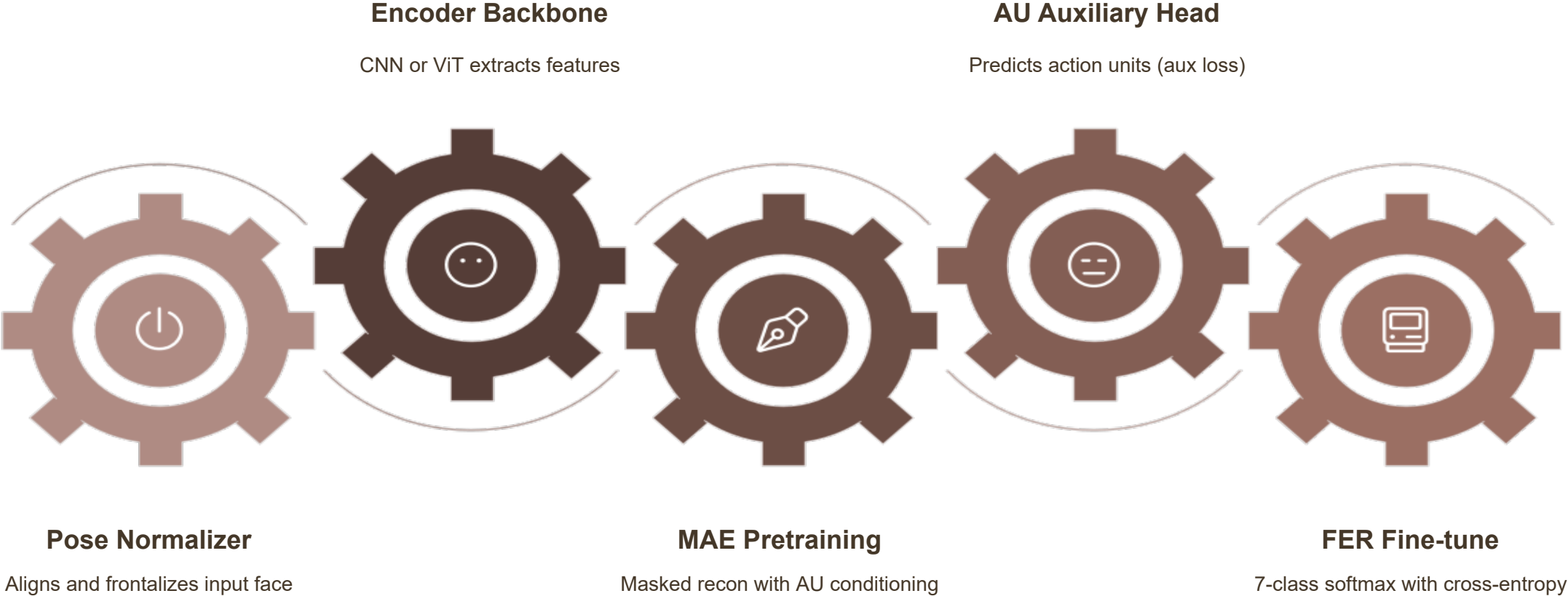
We propose learning rich facial priors by reconstructing masked patches, explicitly conditioned on Action Unit (AU) predictions. This fosters inherent robustness to occlusions and sensitivity to subtle expressions.

Lightweight Pose Normalization

An integrated 3D frontalization module (spatial transformer) will standardize facial poses, significantly increasing consistency and reducing variation in feature extraction, regardless of input angle.

Our work introduces a novel framing: leveraging multi-task AU supervision directly within the self-supervised pretraining phase to guide feature learning.

System Overview: AU-MAE + Pose-Normalized FER



Our system processes an input face image through a **Pose Normalizer**, feeding into an **Encoder** (CNN/ViT backbone). During pretraining, a **Masked Autoencoder** reconstructs occluded patches, crucially **conditioned on AU predictions**. In fine-tuning, a dedicated **FER head (7-class softmax)** and an **AU auxiliary head** collaborate. Optimization involves MAE loss, AU binary cross-entropy, and FER cross-entropy, ensuring robust and interpretable learning.

Scope, Use Cases & Novelty Validation

1) Scope of the System

- Focus: FER in unconstrained, real-world conditions (pose variation, occlusions, limited labels).
- Goal: Robust, interpretable FER pipeline using AU-guided pretraining + pose normalization.

2) Target Use Cases

- Healthcare: Monitor patient emotions in telemedicine sessions.
- HCI: Adaptive interfaces responding to user emotions.
- Robotics: Social robots interpreting facial cues in dynamic environments.

3) Novelty Check

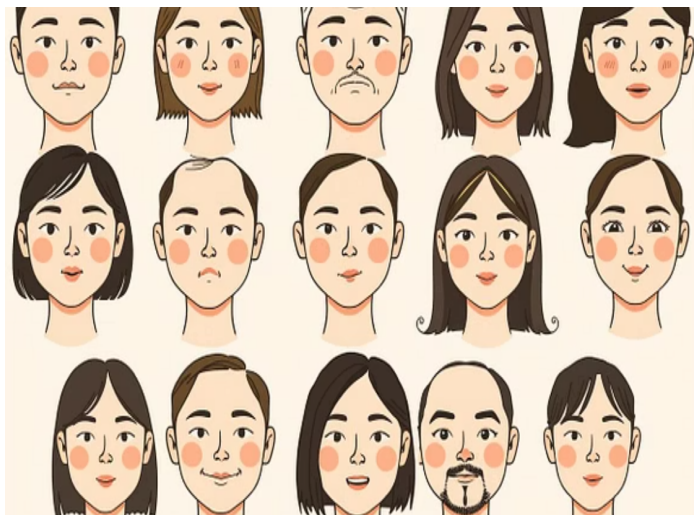
- Searched arXiv, Google Scholar, GitHub with keywords: AU-guided FER, pose-normalized FER, AU-MAE.
- Related work: AU-conditioned learning and pose normalization studied separately.
- Gap: No integrated system combining AU-conditioned self-supervised pretraining with pose frontalization.

Dataset Strategy



Training & Pretraining: AffectNet

Leveraging the extensive AffectNet dataset, including both its labeled and vast unlabeled portions. This provides the necessary scale for effective self-supervised pretraining with AU conditioning.



Validation & Benchmark: RAF-DB, FER-2013

These established datasets will serve as our primary benchmarks for validating the model's generalization capabilities and comparing performance against state-of-the-art methods.



Bonus: "Data-in-the-Wild" Collection

To further assess real-world robustness, we plan to curate a small, challenging dataset from smartphone images, specifically targeting various occlusion types and extreme pose variations.

Rigorous Evaluation Plan

Performance Metrics

- **Accuracy & Balanced Accuracy:** To assess overall performance and handle class imbalance effectively.
- **Macro-F1 Score:** Providing a robust measure across all expression classes.
- **Confusion Matrices:** For detailed insight into per-class prediction strengths and weaknesses.
- **Robustness Analysis:** Quantifying performance degradation across varying pose angles and specific occlusion types.

Comparative Baselines

- **ResNet50 FER Classifier:** Standard ImageNet-pretrained model.
- **Attention-CNN:** Representative of feature-focused CNN architectures.
- **ViT-based FER Model:** State-of-the-art global feature extractor.

Crucial Ablation Studies

- Impact of AU conditioning (with vs. without).
- Contribution of pose normalization (with vs. without).
- Effectiveness of AU-MAE vs. traditional ImageNet pretraining for encoder initialization.

Expected Challenges & Strategic Mitigation

Compute Cost

Challenge: High computational demands for ViT-based MAE pretraining. **Mitigation:** Explore smaller ViT variants or leverage ResNet-based MAE architectures for efficiency.

Class Imbalance

Challenge: Skewed distribution of expression classes in datasets. **Mitigation:** Implement focal loss during training or apply sample reweighting strategies to balance contributions.

AU Label Scarcity

Challenge: Limited availability of dense, accurate AU labels. **Mitigation:** Employ pseudo-labeling techniques or explore weak supervision methods to augment AU annotations.

Ethical Considerations

Challenge: Potential for dataset bias and misuse of FER technology. **Mitigation:** Thoroughly document dataset biases, transparently discuss limitations, and prevent overclaiming capabilities (e.g., distinguishing FER from personality inference).

Project Timeline

Our project is structured to align with the course schedule, ensuring focused progress and timely delivery of each component.

Weeks 0–2

Proposal & Setup: Finalize approach, dataset selection, and GitHub repository configuration.

Weeks 7–9

Fine-tuning & Baselines: Conduct FER training and establish performance baselines.

Weeks 12–13

Finalization: Prepare comprehensive report, demo video, presentation slides, and reflective analysis.

Weeks 3–6

AU-MAE Pretraining: Execute AU-Conditioned MAE pretraining on the AffectNet dataset.

Weeks 10–11

Ablations & Robustness: Perform cross-dataset evaluation, robustness analysis, and test "data-in-the-wild."

Key Deliverables

GitHub Repository

Full code implementation with detailed commits and a comprehensive README for easy setup and reproduction.

Demo Video

A practical demonstration showcasing real-time FER capabilities on diverse sample images and video streams.

Final Report

In-depth documentation covering related work, system design, novelty, experiments, limitations, and future research.

Presentation & Q&A

This presentation, along with an interactive Q&A session for deeper discussion.

Thank You